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Title of the article: *Preferential Killing of Tetraploid Colon Cancer Cells by Targeting the Mitotic Kinase PLK1*

The content extracted is as follows:

Introduction

Tetraploid cells are frequently found in pre-neoplastic lesions of intestinal cancers [13-15].

Inactivation of p53 facilitates the tetraploidisation of cell lines [16, 17].

Tetraploid cells are intrinsically resistant to genotoxic stress mediated by ionising irradiation or anti-cancer chemotherapeutic agents, including platinum compounds and topoisomerase inhibitors [19, 20].

Thus, tetraploid cells might contribute to chemotherapeutic failure.

The development of novel strategies to target tetraploid tumour cells can improve the treatment outcome for CRC patients.

Second, PLK1 can increase the levels of the oncogenic transcription factor MYC in a positive feedback loop. Third, PLK1 overexpression leads to degradation of the tumour suppressor repressor element-1 (RE-1) silencing transcription (REST) in triple negative breast cancer.

Here, we report that PLK1 inhibition leads to the selective destruction of tetraploid cancer cells.

Results

To evaluate the differential impact of PLK1 knockdown on the survival of tetraploid and diploid colon cancer cells, we utilised previously generated 2n and 4n clones derived from parental human colon carcinoma RKO cells [29].

The diploid and tetraploid clones were transfected with an unrelated small interfering RNA (siUNR) as a control and a SMARTpool siPLK1 (siPLK1a), to knockdown PLK1, for up to 72 h (Fig. 1A).

Compared to their diploid counterparts, tetraploid clones were particularly sensitive to PLK1 knockdown.

We performed a proliferation assay using crystal violet and found that tetraploid cells were more sensitive to PLK1 knockdown compared to their diploid counterpart (Fig. 1D-F).

The frequency of (DiOC6(3) low PI low) and (DiOC6(3)low PI high) cells analyzed by flow cytometry was indeed higher in tetraploid comparatively to diploid cells (Fig. 1G-H).

The cell death mediated by PLK1 knockdown in tetraploid cells was also confirmed using human colon carcinoma HCT116 (Supplementary Fig. S1A-B) and sarcoma cells (Supplementary Fig. S2A for all supplemental material see www.cellphysiolbiochem).

Next, we focused on small molecule PLK1inhibitors to assess a potentially alternative treatment option to target PLK1 for cancer therapy.

Since Bi2536 was more specific for PLK1 compared to GSK461364 (Supplementary Fig. S3C), we decided to continue our investigation on tetraploid cells using Bi2536.

The tetraploid cells showed a high^e r sensitivity against Bi2536-treatment after 24 h compared to diploid cells (Fig. 2A-C).

Moreover, using flow cytometry-mediated measurement of PI and DiOC₆(3), tetraploid cells showed a higher sensitivity when treated with 5, 10 or 50 nM Bi2536 compared to diploid cells (Fig. 2E).

Consistently, Bi2536 significantly reduced the clonogenic potential of tetraploid compared to diploid cells, either with continuous 5 nM Bi2536 treatment (75.3 +/- 25 diploid vs 4.3 +/- 3 tetraploid) or after 24 h drug washout (217 +/- 37 diploid vs 148 +/- 35 tetraploid) (Fig. 2F-G).

The preferential toxicity of the Bi2536 to tetraploid cells was also confirmed using the human colon carcinoma HCT116 (Supplementary Fig. S1C-D) and sarcoma (Supplementary Fig. S2B) cells lines. Taken together, these findings demonstrated that targeting PLK1

represents an efficient strategy to preferentially kill tetraploid colon cancer cells.

To characterise the cell death pathway engaged after PLK1 inhibition, tetraploid RK0 colon cancer cells were treated for 72 h with DMSO (as a control) or 10 nM Bi2536.

Moreover, the co-treatment of the cells with the broad-spectrum caspase inhibitor Z-Val-Ala-Asp-fluoromethylketone (Z-VAD-fmk) significantly reduced the death of tetraploid cells that responded to PLK1 inhibition (Fig. 3E).

Further confirming the role of caspases in the execution of apoptosis, tetraploid cells that responded to Bi2536 manifested the apoptosis-associated cleavage of poly (ADP-ribose) polymerase (PARP) (Fig. 3F).

These results indicated that PLK1 inhibition triggers the intrinsic pathway of apoptosis in tetraploid colon cancer cells.

Thus, we investigated the effect of p53 inhibition when PLK1 was knocked down or inhibited by treating the tetraploid RK0 cells with cyclic pifithrin- (PFT-), a pharmacological antagonist of p53 [34], in the presence of siPLK1 (Fig. 3G) and with Bi2536 co-treatment (Fig. 3H).

When analysing cell cycle profiles using flow cytometry after staining fixed cells with the DNA dye PI, we observed that inhibition or depletion of PLK1 perturbed the diploid and tetraploid cell cycle progression.

Some cells underwent mitotic slippage and became hyper-diploid and hyper-tetraploid, respectively.

Notably, the fraction of hyperploidy induced by PLK1 inhibition was higher in tetraploid compared to diploid cells (Fig. 4B).

Indeed, the diploid cells stayed in the G2/M phase longer than tetraploid cells, and approximately 15% of the cells became hyper-diploid after 72 h. However, tetraploid cells executed mitotic slippage earlier; the hyperploidy fraction was approximately 16% at 24 h and reached 32% at 72 h. These findings suggest that the anti-tetraploid effect of PLK1 inhibition came from the relative inability of tetraploid (as compared to diploid) cells to tolerate any further increase in ploidy.

Targeting PLK1 preferentially kills tetraploid colon cancer cells by triggering an uncontrolled and lethal hyperploidy programme, followed by a mitotic catastrophe executed by the intrinsic apoptotic pathway.

PLK1 and tubulin polymerisation inhibitors synergistically induced apoptosis in tetraploid colon cancer cells. Given that PLK1 inhibition induced mitotic catastrophe and polyploidy after a long arrest in the mitotic phase, we asked whether the combination of PLK1 inhibition with a microtubule poison could promote any synergistic effects concerning cells death in tetraploid colon cancer cells.

We found a concentration dependent cell survival in tetraploid cells.

Discussion

In this study, we proposed a novel strategy for targeting tetraploid colon cancer cells based on the abrogation of the mitotic kinase PLK1.

By using tumour cells that display distinct levels of ploidy (diploid and tetraploid), we showed that the depletion or inhibition of PLK1 effectively and preferentially killed tetraploid colon cancer cells via mitotic catastrophe and the activation of a mitochondrial- and caspase dependent apoptosis.

In the Cancer Genome Atlas (TCGA) database (<https://tcgadata.nci.nih.gov/tcga/>), PLK1 messenger RNA (mRNA) expression is significantly higher in tumour compared to normal tissue (across 24 different types of cancers).

In comparison to other cancer types, colorectal cancer exhibits high PLK1 expression [26].

Moreover, a previous study showed that upregulated PLK1 expression correlates with inferior survival outcomes in patients with rectal cancer [37].

PLK1 overexpression induces chromosomal instability [38], a finding that places this kinase as a promising antineoplastic target for cancers with a high level of genomic instability, especially colorectal tumours.

The Bi2536 inhibitor is the most commonly molecule used in the characterisation of PLK1 function and anticancer activity [39].

Moreover, Bi2536 was included in a completed phase II clinical trial for patients with solid tumours, including non-small cell lung, pancreatic or hormone-refractory prostate cancer (clinicaltrials.gov, NCT00376623, NCT00706498, NCT00412880 and NCT00710710).

In our experimental model, either depletion or chemical inhibition of PLK1 provoked apoptosis in both diploid and tetraploid colon cells.

However, tetraploid cells showed higher sensitivity when PLK1 was targeted. Tetraploid cell death was executed via the intrinsic apoptotic pathway, as demonstrated by the loss of mitochondrial transmembrane potential ($\Delta\psi$), accumulation of intracellular calcium, activation of caspase-3 and PARP, DNA degradation, cell scrambling, phosphatidylserine (PS) exposure and, ultimately, the loss of plasma membrane integrity.

What is the mechanism through which PLK1 inhibition is particularly lethal for tetraploid cells?

The tetraploid cells responded similarly, but with completely different kinetics.

It appeared that tetraploid cells death upon PLK1 inhibition was caused by excessive genome reduplication and consequent hyperploidy as well as their incapacity to support accumulation of additional DNA content.

Our findings are consistent with previous studies showing that the high sensitivity of tetraploid cells to the inhibition of mitotic key effectors, especially those with a direct interaction with microtubules.

Inhibition of the protein kinase Aurora B preferentially kills tetraploid compared to diploid hepatocellular carcinoma cells, but it also shows a similar effect on untransformed tetraploid vs diploid mouse fibroblasts.

MPS1 inhibition sensitises the tetraploid colon cancer cells also via the activation of the mitotic catastrophe following a polyploid/aneuploid cascade [46, 47].

CHK1 loss leads to chromosome misalignment and lagging chromosomes and thus preferential killing of tetraploid colon cancer cells [50].

The inhibition/depletion of these proteins also sensitises and preferentially kills tetraploid compared to diploid colon cancer cells after mitotic catastrophe activation [51, 52].

We must emphasise that not all the SAC components or important players of the microtubule dynamics can preferentially sensitise the tetraploid cancer cells.

The inhibition of Aurora A, an important kinase for centrosome maturation, mitotic spindle formation and cytokinesis, does not kill tetraploid hepatocellular carcinoma [42, 53].

We found that co-treatment with a low dose of the PLK1 inhibitor Bi2536 and paclitaxel did not induce any synergistic lethality in tetraploid colon cancer cells.

On the other hand, co-treatment of a low dose of the PLK1 inhibitor Bi2536 and a low dose of either vincristine or colchicine induced strong synergistic toxicity in tetraploid colon cancer cells.

Conclusion

In conclusion, this novel approach to exploit PLK1 inhibitors and microtubule poisons constitutes a promising therapeutic strategy based on mitotic catastrophe for colon cancer.